CAPITAL ASSET PRICING MODEL: APPLICATIONS

Learning Outcomes

- calculate and interpret the expected return of an asset using the CAPM
- describe and demonstrate applications of the CAPM and the SML

The CAPM offers powerful and intuitively appealing predictions about risk and the relationship between risk and return. The CAPM is not only important from a theoretical perspective but is also used extensively in practice. In this section, we will discuss some common applications of the model. When applying these tools to different scenarios, it is important to understand that the CAPM and the SML are functions that give an indication of what the return in the market *should* be, given a certain level of risk. The actual return may be quite different from the expected return.

Applications of the CAPM include estimates of the expected return for capital budgeting, comparison of the actual return of a portfolio or portfolio manager with the CAPM return for performance appraisal, and the analysis of alternate return estimates and the CAPM returns as the basis for security selection. The applications are discussed in more detail in this section.

Estimate of Expected Return

Given an asset's systematic risk, the expected return can be calculated using the CAPM. Recall that the price of an asset is the sum of all future cash flows discounted at the required rate of return, where the discount rate or the required rate of return is commensurate with the asset's risk. The expected rate of return obtained from the CAPM is normally the first estimate that investors use for valuing assets, such as stocks, bonds, real estate, and other similar assets. The required rate of return from the CAPM is also used for capital budgeting and determining the economic feasibility of projects. Again, recall that when computing the net present value of a project, investments and net revenues are considered cash flows and are discounted at the required rate of return. The required rate of return, based on the project's risk, is calculated using the CAPM.

Because risk and return underlie almost all aspects of investment decision making, it is not surprising that the CAPM is used for estimating expected return in many scenarios. Other examples include calculating the cost of capital for regulated companies by regulatory commissions and setting fair insurance premiums. The next example shows an application of the CAPM to capital budgeting.

EXAMPLE 9

Application of the CAPM to Capital Budgeting

GlaxoSmithKline Plc is examining the economic feasibility of developing a new medicine. The initial investment in Year 1 is $500 million. The investment in Year 2 is $200 million. There is a 50 percent chance that the medicine will be developed and will be successful. If that happens, GlaxoSmithKline must spend another $100 million in Year 3, but its income from the project in Year 3 will be $500 million, not including the third-year investment. In Years 4, 5, and 6, it will earn $400 million a year if the medicine is successful. At the end of Year 6, it intends to sell all rights to the medicine for $600 million. If the medicine is unsuccessful, none of GlaxoSmithKline's investments can be salvaged. Assume that the market return is 12 percent, the risk-free rate is 2 percent, and the beta risk of the project is 2.3. All cash flows occur at the end of each year.

Hide Solution

Solution:

There is a 50 percent chance that the cash flows in Years 3–6 will occur. Taking that into account, the expected annual cash flows are:

Year 1: –$500 million (outflow)

Year 2: –$200 million (outflow)

Year 3: 50% of –$100 million (outflow) + 50% of $500 million = $200 million

Year 4: 50% of $400 million = $200 million

Year 5: 50% of $400 million = $200 million

Year 6: 50% of $400 million + 50% of $600 million = $500 million

2. Calculate the expected return.

Hide Solution

Solution:

The expected or required return for the project can be calculated using the CAPM, which is = 0.02 + 2.3 × (0.12 – 0.02) = 0.25.

3. Calculate the net present value.

Hide Solution

Solution:

The net present value is the discounted value of all cash flows:

$$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r_t)^t}$$

$$= \frac{-500}{(1+0.25)} + \frac{-200}{(1+0.25)^2} + \frac{200}{(1+0.25)^3} + \frac{200}{(1+0.25)^4} + \frac{200}{(1+0.25)^5} + \frac{500}{(1+0.25)^6}$$

$$= -400 - 128 + 102.40 + 81.92 + 65.54 + 131.07 = -147.07.$$

Because the net present value is negative (–$147.07 million), the project should not be accepted by GlaxoSmithKline.

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Discussion

What do you want to discuss... (required)

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Practice question is wrong:

$\beta i = \rho i,m \sigma i \sigma m$

The beta is

βi=0.8(0.30)0.15=1.6

The expected return is computed using

E(Ri) = Rf + [E(Rm) – Rf]βi

So, E(Ri) = 0.06 + (0.08-0.06)1.6 = 9.2%.

Answer is A

Created 4 months ago by Gabriel Oluwapelumi Oghughu

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